

Project Title

Secure Question Paper Management System using AAA and CIA security principles.

Objective

The objective of this project is to design and implement a secure Question Paper Management System that

- enables authorized users to create, upload, manage, approve and access question papers while ensuring confidentiality, integrity and availability of sensitive examination data.
- The system employs AAA principles to prevent unauthorized access, maintain accountability, and protect examination material from security threats.
- The system authenticates every user before granting access, authorize operation based on assigned roles and records every critical activity in audit logs.

Stakeholders

Administrator

Manage users, approve questⁿ paper, assign roles, monitor audit logs, backup data.

Faculty / Exam setter

create, upload, edit and submit question papers.

HoD / Exam controller

Review and approve question papers before publication

Student

view or download only officially released question papers.

Act Subject

- Administrator
- Faculty
- HOD
- Students

Objects

- Question Papers
- User Accounts
- Student Records
- Faculty Information
- Audit Log
- Question Paper Repository
- System Configuration
- Database

Controls

- User Authentication
- Role Based Access Control
- Password Hashing
- Activity Tracking

Authentication Mechanism

- Username and Password
- Password Hashing
- Multi-factor Authentication
- CAPTCHA
- Secure Session Management

Authorization Mechanism

Role Based Access Control

- only Admin can create or delete users.
- Faculty can edit only their own question papers.
- student cannot upload or modify papers.
- Only approved question papers are accessible to students.
- Logs are accessible only to Admins.

Accounting

Records every important user activity for accountability and security monitoring.

info recorded:-

- Login time
- Logout time
- Uploaded file
- Downloaded file
- password changes

CIA

Confidentiality

- user authentication
- Role Based Access Control
- Password Hashing

Data Encryption
Secure Login

session

transparency

audit

Integrity

- Approval Workflow
- Audit Logs
- Hash Verification

audit logs

audit logs

Availability

- Fault Tolerance
- Reliable server Infrastructure

Expected outcomes

- secure storage of question papers.
- Elimination of unauthorized access.
- Controlled access through RBAC.
- Reduced risk of question paper leak.
- Better examination management with enhanced security.

Architectural Diagram

